

Voice Cloner - Application Requirements and Acceptance Criteria

ICS 499 Capstone, Group 12 (Mubarek Idris, Mikita Misiulia). Iteration FP4 draft. Repo: github.com/Mubarekidriss/Voice-Cloner (main, after the modular refactor).

1. What the app does

The user types text, records or uploads a short voice sample, and picks a language. The app returns two audio outputs from the same XTTS v2 model and the same text and language: a baseline voice (predefined XTTS speaker "Ana Florence") and a clone of the sample voice. The user compares them side by side. Interface: Gradio web UI. Code modules: config, tts_engine, voice_cloner, ui, app.

2. Functional requirements

ID	Requirement	Acceptance criterion / status
R1	User can enter text to be spoken.	Empty or blank text shows the message "Type some text first." and no audio is generated. Status: in code.
R2	User can record from the microphone or upload an audio file as the voice sample, and play it back before converting.	Both input methods are present in the UI and the sample plays back. No sample shows "Record or upload a voice sample first." Status: in code.
R3	User can pick one of 17 languages (English, Spanish, French, German, Italian, Portuguese, Polish, Turkish, Russian, Dutch, Czech, Arabic, Chinese (Mandarin), Hungarian, Korean, Japanese, Hindi). Default English.	On startup the app checks that the loaded model supports all 17 codes and stops with a clear error if not. English and Spanish clones generate. Russian was verified in a class demo with the instructor (Mikita, week of Sep 29). The other 14 are checked per the FP6 plan (core languages fully, rest spot-checked). Status: list in code; English and Spanish verified on Colab; Russian verified in class demo.
R4	App generates a baseline output and a cloned output from the same model, text and language.	Both audio players are filled after one click on Submit, and both play. Verified on Colab T4 for English and Spanish (about 4 s each).
R5	App loads XTTS v2 once at startup and reports whether it runs on GPU or CPU.	Startup prints the device. On CPU it prints a slow-speed warning. Model load failure ends with an error message, not a blank UI.
R6	Temporary audio files are cleaned up if a conversion fails.	After a forced failure, no leftover temp .wav files from that run remain. Status: cleanup is in code; test to be written (FP9).
R7	Responsible-use notice is visible in the UI.	The notice about only cloning a voice with permission appears on the page. FP8 adds consent messaging, retention and privacy.
R8	Audio input formats: the app accepts the common audio formats named in the FP5 plan.	Exact list to be confirmed with the FP5 plan. Acceptance: for each listed format, upload one file and get both outputs with no error, on Windows and Mac. Today Gradio converts the input to WAV and static-ffmpeg supplies ffmpeg/ffprobe.

3. Supported platforms

- Operating systems: Windows and macOS (local run with VS Code), and Google Colab (Linux, T4 GPU) as the recommended demo path. Acceptance: install from requirements.txt and run the app on each, then generate one English clone.
- Browsers (proposed, to confirm with Mikita): current Chrome, Edge, Firefox and Safari. Acceptance: open the app page, record or upload a sample, generate both outputs, play both. Microphone recording needs a secure context (localhost or https).
- Hardware: a GPU is strongly recommended. CPU works but is very slow (the code warns about this). First run downloads about 2 GB of model weights.
- Pinned packages: coqui-tts 0.27.5, torch 2.8.0, torchaudio 2.8.0, transformers 4.57.6, gradio, static-ffmpeg 3.0. Known issue: on Colab, uninstall the preinstalled torchvision to avoid a torch version clash.
- Run command (local): python modules/app.py from the repo root, per the README.

4. Out of scope for now (planned in later iterations)

Separate XTTS backend and hardening (FP6), full UI with compare and download (FP7), consent, retention and privacy (FP8), automated unit and integration tests (FP9), https deployment and clean install (FP10), final docs and v1.0 freeze (FP11).

5. Open items

- README still describes the standard voice as gTTS; the code now uses the XTTS default speaker. README needs updating.
- Confirm the exact audio format list and browser list with Mikita and the FP5 plan.
- Module ownership for FP5-FP7 is still to be agreed.